Sets, Classes, and the First Warnings About Size

N-20210204

§1.1 Why classes enter before functions

The basic language of sets, classes, indexed families, and functions begins with a distinction that is easy to underestimate: sets may be elements of other collections, whereas proper classes cannot be treated in the same way without recreating the classical paradoxes.

Definition 1.1.1. A **set** is a class which can be an element of another class. A class which is not a set is called a **proper class**.

This is not meant to be the whole foundation of set theory, but it explains the working convention of the note: some collections are small enough to be members of other collections, while others are too large and must remain proper classes.

> **Caution**
>
> The distinction between sets and proper classes is formally essential: it is one of the mechanisms that prevents the usual paradoxes. In a class theory such as NBG, sets are precisely the classes that may occur as elements of classes; proper classes may not be elements.

§1.2 Russell's class as the first warning

The first serious example is Russell's construction. Consider

$$M = \{X \mid X \text{ is a set and } X \notin X\}.$$

At first sight this is a natural class: many sets are not members of themselves. For instance, the set of all books is not itself a book.

> **Proposition 1.2.1**
>
> The class M is not a set.

Proof. Assume, for contradiction, that M is a set. Then it makes sense to ask whether $M \in M$. By the defining condition of M,

$$M \in M \quad \Longleftrightarrow \quad M \notin M,$$

which is impossible. Hence M must be a proper class. □

This is the first place where the note is not merely listing definitions. The example shows why unrestricted comprehension cannot simply produce a set from every condition.

§1.3 Power sets and indexed families

Definition 1.3.1. The **power set axiom** asserts that for every set A, the class

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

of all subsets of A is itself a set. It is also common to write 2^A for this power set.

The next construction is an indexed family. Given a family of classes or sets

$$\{A_i \mid i \in I\},$$

its union and intersection are written

$$\bigcup_{i\in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\},$$

and

$$\bigcap_{i\in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}.$$

Remark 1.3.2 — The indexing collection I need not be finite or countable. This is a useful mental shift: the notation $\{A_i\}_{i\in I}$ is not just a long list. It may be a genuinely infinite family.

Caution

If I is allowed to be a proper class, then extra care is needed. In ordinary ZFC, one usually speaks of set-indexed families. In a class-theoretic setting, class-indexed families can be discussed, but one must check separately when their unions or intersections are sets.

If A and B are classes, the relative complement of A in B is

$$B \setminus A = \{x \mid x \in B \text{ and } x \notin A\}.$$

If all classes under discussion are subsets of a fixed universe U, then $U \setminus A$ is often denoted A^c.

§1.4 Set identities

The standard set identities take the following form:

Proposition 1.4.1

Let A, B, and B_i be subsets of a fixed universe U. Then

$$A \cap \left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I}(A \cap B_i),$$

$$A \cup \left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I}(A \cup B_i),$$

$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c, \qquad \left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c.$$

Moreover, $A \cup B = B \cup A$, $A \subseteq B$ means every element of A lies in B, and $A \cap B = A$ when $A \subseteq B$.

Proof. All of these follow by chasing membership. For instance, an element x lies in $A \cap (\bigcup_i B_i)$ exactly when $x \in A$ and $x \in B_i$ for at least one i, which is exactly the condition that $x \in \bigcup_i (A \cap B_i)$. The De Morgan laws are proved similarly after replacing "there exists" by "for all" under negation. $\square$

§1.5 Images, inverse images, and functions

Let $f : A \to B$ be a function. If $S \subseteq A$, the image of S under f is

$$f(S) = \{b \in B \mid b = f(a) \text{ for some } a \in S\}.$$

If $T \subseteq B$, the inverse image of T is

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}.$$

Example 1.5.1

For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, we have

$$f(\mathbb{R}) = [0, \infty), \qquad f^{-1}(\{9\}) = \{-3, 3\}.$$

A map $f : A \to B$ is **injective** if $f(a) = f(a')$ implies $a = a'$. It is **surjective** if $f(A) = B$. It is **bijective** if it is both injective and surjective.

Proposition 1.5.2

For composable maps $A \xrightarrow{f} B \xrightarrow{g} C$:

(i) if f and g are injective, then $g \circ f$ is injective;

(ii) if f and g are surjective, then $g \circ f$ is surjective;

(iii) if $g \circ f$ is injective, then f is injective;

(iv) if $g \circ f$ is surjective, then g is surjective.

§1.6 One-sided inverses

One-sided inverses recognize injective and surjective maps.

Theorem 1.6.1

Let $f : A \to B$ be a function and assume $A \neq \varnothing$.

(i) f is injective if and only if there is a map $g : B \to A$ such that $g \circ f = 1_A$.

(ii) f is surjective if and only if there is a map $h : B \to A$ such that $f \circ h = 1_B$, provided the required choices of elements from the fibres of f can be made.

Proof. For (i), suppose f is injective. For $b \in f(A)$, there is a unique $a \in A$ with $f(a) = b$. Choose once and for all an element $a_0 \in A$. Define

$$g(b) = \begin{cases} a, & b \in f(A) \text{ and } f(a) = b, \\ a_0, & b \notin f(A). \end{cases}$$

Then $g(f(a)) = a$ for all $a \in A$. Conversely, if $g \circ f = 1_A$, then $f(a) = f(a')$ implies

$$a = g(f(a)) = g(f(a')) = a',$$

so f is injective.

For (ii), if h exists and $f \circ h = 1_B$, then every $b \in B$ equals $f(h(b))$, so f is surjective. Conversely, if f is surjective, each fibre $f^{-1}(b)$ is nonempty. Choosing $h(b) \in f^{-1}(b)$ gives a right inverse. □

Caution

The last choice of one element from each fibre is exactly where the Axiom of Choice may enter. This point will reappear in the chapter on choice and cardinal numbers.

§1.7 A right inverse hides a choice function

Consider a surjection

$$f : \{a, b, c\} \longrightarrow \{r, s\}, \qquad f(a) = f(b) = r, \quad f(c) = s.$$

There are two possible right inverses: the image of r may be either a or b, while the image of s must be c. A right inverse does exactly what its formula says: it chooses one representative from every nonempty fibre.

For a general surjection $f : A \to B$, those fibres form a family

$$(f^{-1}(\{b\}))_{b \in B}.$$

A section $s : B \to A$ satisfying $f \circ s = 1_B$ is precisely a simultaneous choice of one element from each member of that family. For a finite B, finitely many individual choices suffice. For an arbitrary family, this is where a set-theoretic axiom enters.

Theorem 1.7.1 (A useful equivalent form of Choice)

Over ZF set theory, the Axiom of Choice is equivalent to the assertion that every surjection of sets admits a right inverse.

Proof. If Choice holds and $f : A \to B$ is onto, choose $s(b) \in f^{-1}(\{b\})$ for every b. Then $f(s(b)) = b$.

Conversely, take any indexed family $(X_i)_{i \in I}$ of nonempty sets. Its disjoint union

$$E = \coprod_{i \in I} X_i = \{(i, x) : i \in I,\ x \in X_i\}$$

has a natural surjection $\pi : E \to I$, $(i, x) \mapsto i$. By hypothesis it has a section $s : I \to E$. Writing $s(i) = (i, c(i))$ defines a choice function $c(i) \in X_i$. The assertion for all surjections therefore supplies a choice function for every such family. □

It is the same elementary fibre picture in both directions. The subtlety is that constructing a right inverse for one explicitly given finite function is easy, whereas asserting their existence for *all* surjections is already the full Axiom of Choice.